MODULE 2 · LECTURE 12-a

CS 6530 Applied Cryptography

Mathematical Foundations

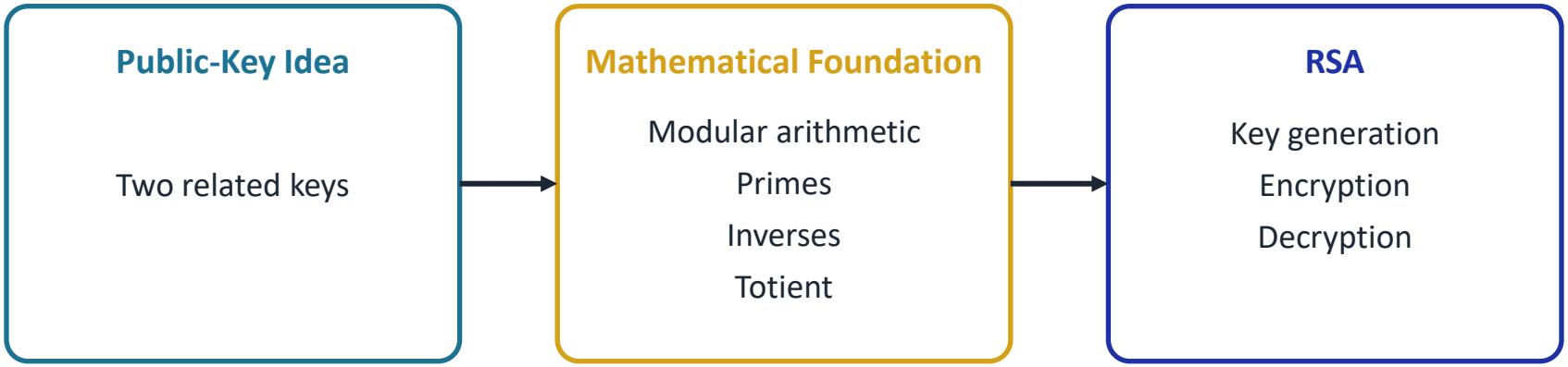
Module 3 – Public-Key Cryptography

From the Public-Key Idea to RSA

Public-key cryptography uses two mathematically related keys:

Public Key
Shared with others

Private Key
Kept secret by its owner



RSA constructs its public/private key relationship using properties of integers and modular arithmetic.

Knowing the public key must not make recovery of the private key computationally feasible.

Key idea: The mathematics provides the mechanism that connects RSA's public and private keys.

Modular Arithmetic — Working with Remainders

Two integers are congruent modulo n when their difference is divisible by n.

$$a \equiv b \pmod{n} \iff n \mid (a - b)$$

Example

$$17 \equiv 5 \pmod{12}$$

because 17 − 5 = 12 is divisible by 12.

Likewise, $29 \equiv 5 \pmod{12}$.

Reduce While Computing

$$(17 \times 11) \bmod 12$$

$17 \equiv 5, \quad 11 \equiv -1 \pmod{12}$

Therefore:

$17 \times 11 \equiv 5(-1) \equiv -5 \pmod{12}$

$$17 \times 11 \equiv 7 \pmod{12}$$

Key idea: Reduce intermediate values modulo n; the final modular result is preserved.

Modular Multiplicative Inverse

The modular inverse of a modulo n is a value that multiplies with a to give 1 modulo n.

$$ax \equiv 1 \pmod{n}$$

$$x \equiv a^{-1} \pmod{n}$$

Worked Example — Find $3^{-1} \pmod{10}$

We need: $3x \equiv 1 \pmod{10}$.

For $x = 7$: $3 \times 7 = 21 \equiv 1 \pmod{10}$.

$$3^{-1} \equiv 7 \pmod{10}$$

Does an inverse always exist? — No.

A modular inverse exists if and only if $\gcd(a, n) = 1$.

Example: 2 has no inverse modulo 10 because gcd(2, 10) = 2.

Why This Matters for RSA

$$\gcd(e, \phi(n)) = 1 \implies d = e^{-1} \pmod{\phi(n)}$$

Key idea: Coprimality guarantees that the modular multiplicative inverse exists.

Prime Numbers and Coprimality

Prime Numbers

A prime number $p > 1$ has exactly two positive divisors:

$$1 \quad \text{and} \quad p$$

Examples:

$$2,\ 3,\ 5,\ 7,\ 11,\ 13,\ 17,\ \ldots$$

A composite number has additional factors:

$$15 = 3 \times 5$$

Coprimality

The greatest common divisor, gcd(a, b), is the largest positive integer dividing both a and b.

Two integers are coprime when:

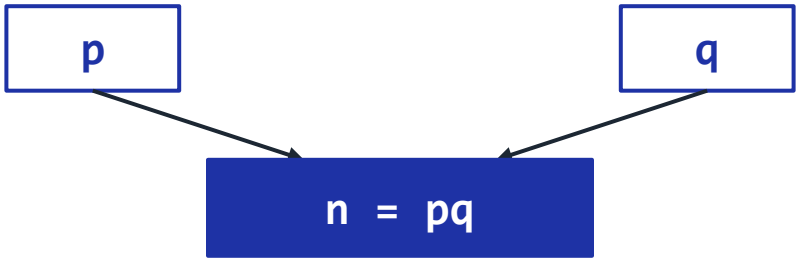
$$\gcd(a, b) = 1$$

$\gcd(7, 20) = 1$ ✓ coprime

$\gcd(6, 20) = 2$ ✗ not coprime

RSA Connection

RSA begins with two distinct large primes:



The key generator knows the factorization p, q; someone given only n does not automatically know it.

Key idea: RSA builds its modulus from two distinct primes whose factorization is known to the key generator.

Euler's Totient Function

Euler's totient function $\phi(n)$ counts the positive integers less than n that are coprime to n.

$$\phi(n) = |\{k : 1 \le k < n, \gcd(k, n) = 1\}|$$

Worked Example — $\phi(10)$

Numbers less than 10:

1	2	3	4	5	6	7	8	9

Only 1, 3, 7, 9 are coprime to 10.

$$\phi(10) = 4$$

RSA-Relevant Results

For a prime p:

$$\phi(p) = p - 1$$

For distinct primes p, q, n=pq:

$$\phi(pq) = (p-1)(q-1)$$

Example: $p = 5,\ q = 11$

$$n = 55, \quad \phi(55) = 4 \times 10 = 40$$

Key idea: If the prime factors p, q are known, computing φ(n) for n = pq is straightforward.

Euler's Theorem — The Mathematical Cycle

For integers a and n, if:

$$\gcd(a, n) = 1$$

then Euler's theorem states:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

Worked Example

Take $a = 3$, $n = 10$. From the previous slide: $\phi(10) = 4$.

Since $\gcd(3, 10) = 1$, Euler's theorem gives: $3^4 \equiv 1 \pmod{10}$.

$$3^4 = 81, \quad 81 \bmod 10 = 1$$

Why RSA Needs This

RSA constructs its exponents so that $ed = 1 + k\phi(n)$ for some integer k.

This, with Euler's theorem, will later explain why: Encrypt → Decrypt → Original Message

Key idea: Powers separated by multiples of φ(n) exhibit a modular structure that RSA exploits.

Efficient Modular Exponentiation

RSA repeatedly needs to compute $a^e \bmod n$. *Computing the enormous integer a^e first is unnecessary.*

Instead leverage the math property .(a·b) mod n = [(a mod n)(b mod n)] mod n.

Square → Reduce → Square → Reduce

Compute $7^{13} \bmod 11$

First express the exponent: $13 = 8 + 4 + 1$

Power	Calculation	Modulo 11
7^1	7	7
7^2	$7^2 = 49$	5
7^4	$5^2 = 25$	3
7^8	$3^2 = 9$	9

Now: $7^{13} = 7^8 \cdot 7^4 \cdot 7$

Therefore: $7^{13} \bmod 11 \equiv 9 \cdot 3 \cdot 7 \bmod 11$

Reduce: $9 \times 3 = 27 \equiv 5 \pmod{11}$ → $5 \times 7 = 35 \equiv 2 \pmod{11}$

$$7^{13} \equiv 2 \pmod{11}$$

Repeated multiplication scales with the exponent e.
Square-and-multiply requires only **O(log e)** modular squarings/multiplications.

Key idea: Modular exponentiation makes very large exponents computationally manageable without constructing a^e.